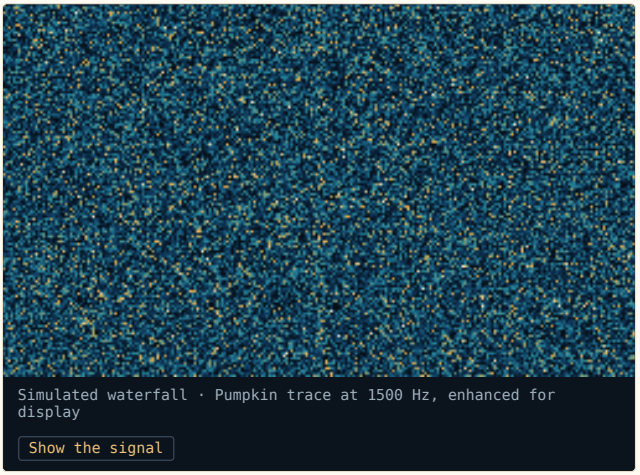
PUMPKIN · CUCUMBER · CHAT · 220 MHZ

−30.8 dB

Weak-signal beacons and chat for every VHF radio. Testers wanted worldwide.

Uranus is free beacon and chat software for the 2 m, 1.25 m and 70 cm amateur bands. Its coherent mode, **Pumpkin**, decodes a one-minute transmission 30.8 dB below the noise. Its FM mode, **Cucumber**, gives any FM radio a weak-signal beacon and chat. Now it needs stations on the air.



Simulated waterfall · Pumpkin trace at 1500 Hz, enhanced for display

Show the signal

BRIDGETOWN, BARBADOS — 8 November 2026. The Uranus project, led by Kammie Holder, 8P5AB, today released version 0.7.0 of its free beacon software for the 2 m and 70 cm amateur radio bands. The release introduces **Pumpkin**, a coherent weak-signal mode that recovers its carrier from known pilot symbols and demodulates it coherently.

In full audio testing, where the decoder must find the signal's timing and frequency on its own, Pumpkin-60 decoded at −30.8 dB and Pumpkin-15 at −24.7 dB. That is 2.7 to 4.0 dB better than WSJT-X's FST4 at the same slot length and the same transmitter peak power.

That puts Pumpkin ahead of the published FT8, FST4 and Q65 figures for the same slot length, whenever the receiving station has no prior knowledge of the caller.

Uranus is freeware, and the project's next step depends on other amateurs. Every result so far comes from simulation and audio testing. The team is inviting stations around the world to put Pumpkin on the air, report what they decode and help turn laboratory numbers into field results.

The design rules came out of the measurements. An early filtered-BPSK version lost 3.2 dB to its peak-to-average ratio. The team switched to MSK, which keeps a constant envelope, so a transceiver runs it at full power with no loss of sensitivity.

Uranus runs on a Raspberry Pi or a Windows PC with any Hamlib-controlled all-mode rig, and also sequences APRS, FT8 and WSPR beacons.

> “We set out to decode below −30 dB in under a minute, and in testing Uranus did it on a 2-metre beacon here in Barbados. Uranus is free for everyone. Now we need stations on the air to prove it, and we invite amateurs around the world to join our on-air tests.”
>
> **KAMMIE HOLDER, 8P5AB** · URANUS PROJECT LEAD

Five design choices behind Pumpkin

Each one was kept because it measured better, not because it sounded good.

01

CHOICE

Trade speed for sensitivity.

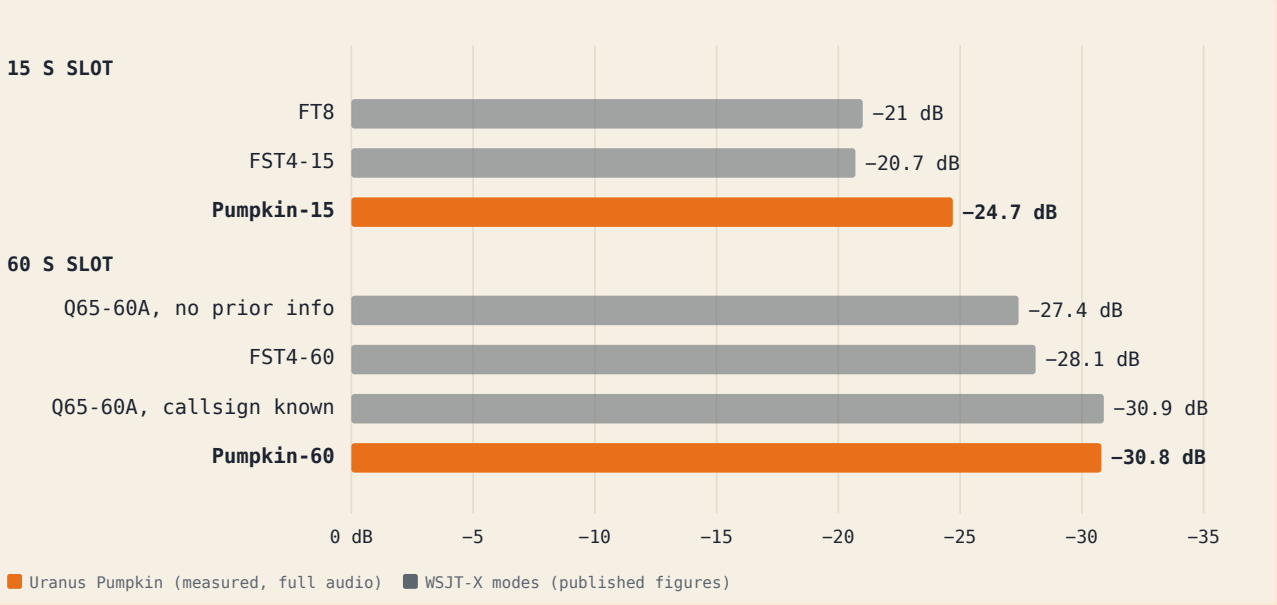
WHAT IT GIVES

A ladder of submodes from 15 s to 300 s. Every halving of rate gains about 3 dB, and Pumpkin-60 occupies under 10 Hz.

02	CHOICE A built-in carrier reference.	WHAT IT GIVES One known pilot in every five symbols, tracked by a Kalman filter and smoother that also uses the pilots that come after each symbol.
03	CHOICE Coherent detection at full transmitter power.	WHAT IT GIVES Coherent MSK: BPSK sensitivity with a constant envelope (0.0 dB peak-to-average, measured).
04	CHOICE Strong error correction.	WHAT IT GIVES A K=9, rate ⅓ convolutional code with a soft-decision Viterbi decoder, plus a CRC-16 to reject false decodes.
05	CHOICE Precise frequency at both ends.	WHAT IT GIVES GPS-disciplined oscillators, so the decoder searches only ±0.6 Hz for Pumpkin-60.

The numbers

Signal strength needed for 50 % of transmissions to decode, SNR in 2500 Hz (the WSJT-X convention). Longer bars mean weaker signals can be decoded.



What's in v0.7.0

Band and mode sequencer

2 m, then 1.25 m, then 70 cm, with a 3-minute pause between bands. Each slot is aligned to its UTC boundary, with 1 minute between transmissions.

Pumpkin coherent mode

Pilot-aided carrier recovery, constant-envelope MSK and a soft Viterbi decoder. Five submodes from 13 s to 4.4 min on air.

Cucumber for FM radios

The Pumpkin signal sent as audio through any FM radio, with no GPS oscillator. In simulation it decodes more than 20 dB below where FM voice and APRS stop.

Radios and SDRs from a list

Pick from 19 radio and SDR profiles, with hover help on every setting. Receive with an RTL-SDR; transmit with a HackRF One (experimental, filter required).

Chat at the edge of the noise

Type messages over Pumpkin or Cucumber, about 11 characters per transmission, taking turns like FT8, with automatic callsign ID.

Live station map and propagation

Every station by band (2 m, 1.25 m, 70 cm) with distance, bearing and SNR, ringed blue when online and red when not. Each contact gets an estimate of the propagation that carried it, with the reasons.

PSK Reporter uploads

Listen mode decodes Pumpkin every slot and can upload each decode to PSK Reporter, so Pumpkin spots appear alongside other modes worldwide.

Pi, Linux and Windows

Ready-made downloads for all three: Uranus.exe for Windows, plus one-command installers for Raspberry Pi and Linux.

Listen before transmit

APRS waits for a clear channel using the rig's S-meter, so the beacon does not transmit over other stations.

Where Pumpkin stands

MODE	SLOT	THRESHOLD	STRENGTH	TRADE-OFF
Pumpkin-60 (Uranus)	52.8 s	−30.8 dB	Best single-transmission result here without prior knowledge of the caller	Needs a GPSDO at both ends; best on stable paths
Q65-60A (WSJT-X)	60 s	−27.4 / −30.9 dB	Built for scatter and EME fading; reaches −30.9 dB when the caller's callsign is already known	Needs prior information to match Pumpkin
FST4-60 (WSJT-X)	51.8 s	−28.1 dB	Proven on air, widely decoded	2.7 dB less sensitive than Pumpkin-60
WSPR	110.6 s	−31.4 dB	Thousands of receivers report to WSPRnet	Takes twice as long on air

What Pumpkin and Cucumber can be used for

Each transmission carries a callsign, a grid square and a short telemetry value, or 11 characters of text. That is enough for a lot.

Propagation beacons

Round-the-clock beacons on 2 m, 1.25 m and 70 cm that show exactly when and where a band opens, with every spot on the map and on PSK Reporter.

PUMPKIN · CUCUMBER

A ducting watch on 220 MHz

A network of ordinary FM radios that detects tropospheric ducting across the Caribbean long before voice contacts become possible.

CUCUMBER

Messages when voice fails

Short text between licensed operators at the edge of range during storms and outages, at signal levels where voice is unreadable. A supplement to established emergency nets, not a replacement.

CHAT

Remote telemetry

A 14-bit value in every beacon: battery voltage, temperature or wind from a solar-powered station on a hilltop, or a high-altitude balloon.

PUMPKIN · CUCUMBER

Clubs and schools

Free, open software for teaching signal processing, coding and propagation with real radios, from a phone-charger-powered Pi upwards.

ALL MODES

Citizen science

Every spot carries distance, SNR, fading and a propagation estimate: a growing open dataset for studying VHF and UHF propagation.

PUMPKIN · MAP · PSK REPORTER

NEW IN V0.7.0

Chat at the edge of the noise

Uranus now carries keyboard chat on both modes. Messages are split into 11-character frames, sent in turn like FT8 and joined back together at the other end, with your callsign sent automatically at least every 10 minutes. Text is plain, as amateur rules require.

Bandwidth is no obstacle: Pumpkin occupies under 40 Hz and Cucumber one ordinary 12.5 kHz FM channel. Speed is the trade-off for sensitivity: about 44 characters a minute on the

```
10:00:15 heard 8P5AB GK03
10:01:30 8P5AB: DUCT OPEN ON 220. HEARING J68 AT -24 DB. QSL?
10:02:15 me: QSL 8P5AB. 73 FROM FK93
example log · Cucumber-15
```

15-second submodes, about 11 on the more sensitive 60-second ones.

WEAK-SIGNAL MODE FOR FM RADIOS

Cucumber

Most radios on 1.25 m, and millions on 2 m and 70 cm, are FM only. Cucumber gives every one of them a weak-signal beacon mode. It sends the Pumpkin signal as audio through the radio's data port, and the decoder works from the receiver's audio.

ON AN ORDINARY FM RADIO	ON AIR	DECODES DOWN TO	NEEDS
FM voice or APRS	–	about +10 dB	A clear signal
Cucumber-15	13.2 s	about −11 dB	Squelch open, data-port audio
Cucumber-60	52.8 s	about −13 dB	Squelch open, data-port audio

Carrier-to-noise in a 12.5 kHz FM channel, from simulation with 2.5 kHz deviation and a ±50 ppm sound-card clock. No GPS oscillator is needed: FM returns the audio frequency exactly as it was sent. Cucumber is still about 25 dB less sensitive than Pumpkin on SSB, because FM gives away weak-signal headroom.

The 220 MHz frontier

222

MHZ · 1.25 M BAND · THE AMERICAS ONLY

- **A band most of the world cannot use.** The 1.25 m band is allocated to amateurs only in the Americas (ITU Region 2), and Barbados allocates 222 to 225 MHz.
- **Quiet by nature.** Operators describe the qualities of 2 m with less interference from consumer electronics and a lower noise floor, exactly what a weak-signal mode needs.
- **Waiting to be explored.** Few stations are active and few radios cover it. Uranus runs Pumpkin on 222.130 MHz through a transverter or an SDR, Cucumber on 223.420 MHz through any FM radio, and maps every 1.25 m contact in its own colour.
- **Pioneers wanted.** With Cucumber, even a tri-band FM handheld can join. Your reports will help map this band across the Caribbean.

TESTERS WANTED, WORLDWIDE

“Let’s push the boundaries of past discoveries. The future is today, and tomorrow’s skills must be embraced today.”

KAMMIE HOLDER, 8P5AB · 2026

Uranus is free. The only price is a report of what you heard. Any licensed amateur, anywhere, can take part.

What you need

Any 2 m, 1.25 m or 70 cm radio with a PC or Raspberry Pi. An FM radio is enough for Cucumber; for Pumpkin, an all-mode rig with a GPS-disciplined oscillator. Listeners with only an RTL-SDR dongle are welcome too.

What to do

Run Pumpkin-60 alongside FST4W-120 or Q65-60A on the same path, or Cucumber alongside FM voice, so each mode can be compared like for like.

How to report

Turn on PSK Reporter uploads so your decodes appear on the map automatically, and open an issue on the GitHub page with your path, the decode SNR of each mode and any recordings.

Report a test

Get Uranus

Free for every amateur. From the release page: `Uranus-Windows-exe.zip` (double-click `Uranus.exe`), `Uranus-RaspberryPi.zip` or `Uranus-Linux.zip` (run the installer inside).

Download the latest release | Source on GitHub

Media and technical contact: **Kammie Holder, 8P5AB**, Barbados · via github.com/BajanBob/Uranus

About these results

- **Not yet tested on the air.** All thresholds come from simulation and audio loopback in white Gaussian noise. On-air trials between GPSDO-equipped stations are the next milestone.
- Pumpkin-120 and Pumpkin-300 figures (−34.1 and −38.1 dB) are simulation results; Pumpkin-15 and Pumpkin-60 were measured through the full audio decoder.
- WSJT-X thresholds are the published figures from the FST4 and Q65 quick-start guides. WSJT-X is the work of K1JT and the WSJT development group.
- Operators must check their licence conditions for new digital modes, unattended operation and every frequency they enable.

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